Teaching to the Test: Detecting Test Gaming by AI Agents in Autonomous Scientific Discovery

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Abstract. Visible tests can reward code that reproduces expected answers without performing the requested scientific analysis correctly. We investigate this risk across six scientific analysis tasks and 190 valid agent runs, comparing hidden unit tests, metamorphic checks, and agent self-reviews. Preliminary results show that checks anchored to the supplied dataset expose shortcuts missed by other hidden checks. Agents also acknowledge shortcuts more consistently during review than in their final answers. These observations indicate that passing visible tests and receiving a confident summary are insufficient evidence of correctness, motivating independent checks of scientific code and its reported results.

Keywords: Software testing · AI agents · Metamorphic testing · Mutation testing · Scientific software

1 Introduction

This template is provided for the Research Oriented Term Project (Option 2) for the CIS*4150 Software Reliability and Testing course. It is based on the Springer Nature template, commonly used in academic journals and conferences. The template outlines how to structure your paper, include figures and tables, and cite references from the bibliography.

Once you have reviewed this document, replace the placeholder content with your own in the appropriate sections. I hope you find this helpful!

Kind regards,
Ed

The Introduction sets the stage for the research by providing a high-level overview of background information, defining the problem or research question, and explaining the study's objectives. It typically includes a couple key relevant papers to highlight gaps in current knowledge, justifies the need for the study, and outlines the paper's structure. The introduction should engage the reader by clearly explaining why the research is important and what the study aims to contribute to the field. It often ends with a brief summary of the research approach or hypothesis.

The remainder of this paper is structured as follows: The next section offers a background ... . In Section 3, we outline the methods ..... The findings from our study are presented in Section 4. Section 5 reflects on these findings within a broader context, and Section 6 concludes the paper .....

2 Background

Replace all this with a relevant comprehensive literaure review in your specific area of interest in Software Testing. The Background section provides context and detailed information needed to understand the research problem. It typically

includes a review of existing literature, key concepts, theories, and previous studies relevant to the topic. The goal is to frame the research by outlining what is already known, identifying gaps in the knowledge, and explaining why the current study is necessary. The background helps the reader understand how the research fits into the broader field and what it aims to contribute.

2.1 A Subsection

This is a subsection. Here is a citation [1] and here is another [2]. Notice the non-breaking space (i.e., ~) and the "cite" command. Notice also the quotations.

3 Methodology

Replace all this with your specific methodology that you used in your Software Testing research project. The Methodology section explains how the research was conducted in detail. It provides a clear and precise description of the study design, data collection methods, tools or instruments used, and the procedures followed. This section also outlines the sampling techniques, participants or data sources, and any statistical or analytical methods applied. The goal of the methodology is to allow other researchers to replicate the study or understand how the results were derived. It should be detailed enough to ensure transparency and reproducibility while justifying why certain methods were chosen.

3.1 System Usablity Scale

This section just shows that you can include figures very easily. You first talk about the figure, then it follows. For example, Figure 1 presents the System Usablity Scale that has been used in thousands of usabilty studies. If you create figures, make sure to store them in the "figures" subfolder. This will help keep everything in your LaTeXproject more organized.

4 Findings

Replace all this with the findings (analysis and results) from your research project. The Findings section presents the results of the research without interpretation or analysis. It includes the key data or outcomes derived from the methods used, such as statistical results, experimental observations, or case study outcomes. The findings are often displayed through tables, graphs, or charts to clearly communicate the results. The goal is to objectively present the evidence generated by the research, allowing the reader to see what was discovered before the interpretation is provided in the discussion section.

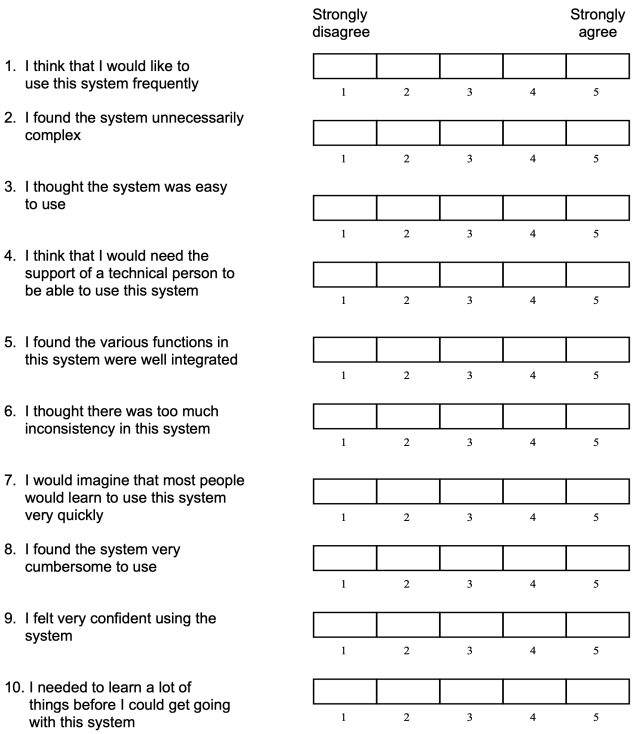


Fig. 1. System Usability Scale

5 Discussion

Replace this section with your Discussion. The Discussion section of a paper should interpret and explain the significance of the research findings. It connects the results to the research questions or hypotheses, explores their implications, and situates them within the broader context of the field. It also compares the findings with previous studies, highlights the study's strengths and limitations, and suggests possible future research directions. The goal is to provide a clear understanding of the relevance and impact of the results.

6 Conclusion

Replace all this with your Conclusion. The Conclusion provides a concise summary of the main findings and their implications. It reinforces the key takeaways from the research, emphasizing the overall contribution to the field. The conclusion often includes final thoughts on the study's significance, practical applications, and, if applicable, recommendations for future research. It should leave the reader with a clear understanding of the study's value and outcomes without introducing new information.

6.1 Future Research

This section discusses future research ...

References

1. Karar, M.E., Shehata, H.I., Reyad, O.: A survey of iot-based fall detection for aiding elderly care: sensors, methods, challenges and future trends. Applied Sciences **12**(7), 3276 (2022)
2. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, Ł., Polosukhin, I.: Attention is all you need. In: Advances in neural information processing systems. pp. 5998–6008 (2017)

Appendix A

Academic Integrity Statement – AI use declaration

Project option	☐ Option 1 Applied Research ☐ Option 2 Research Oriented
Group members	
Final document	Report or paper filename
AI use status	☐ Section 3 limited use ☐ Project-specific use approved in the goal document

Table 1:

Final project check	Confirmation or evidence
Goal document, draft, and final submission were authored by the group within Section 3 permissions	
Sources were read, verified, and cited in the required format	
Code, tests, figures, tables, data, results, and supporting files are genuine	
Contribution statements match the version and repository history	
The required academic integrity page is included and signed	
Project-specific AI use is declared; substantive interactions are documented	

Group declaration

By signing below, we confirm that all project-specific AI use was disclosed and permitted under Section 3. The group authored the submitted work, and every member can explain it. Electronic signatures are acceptable.

Member 1 name, signature, and date	
Member 2 name, signature, and date	
Member 3 name, signature, and date	

Appendix B

Provide evidence detailing whether the contribution among group members was equally divided or not. This should include:

- A complete list of collaborators, accompanied by a brief summary of each member's specific contributions (e.g., one member analyzed the problem, another conducted research on a specific aspect, and all members participated in developing and testing the solution).
- Supporting evidence, such as GitHub or Bitbucket commits, Microsoft Word or Google Docs revision tracking, or similar documentation.

If contributions were not equally distributed, propose a weighted evaluation of each member's contributions. All group members must agree on this distribution and sign the document. For example:

Proposed Weighted Contribution:

Member 1: Tom Smith — Contributed **40%** to this project.

Signature: _____ **Date:** _____

Member 2: John Appleseed — Contributed **40%** to this project.

Signature: _____ **Date:** _____

Member 3: Suzy Banana — Contributed **20%** to this project.

Signature: _____ **Date:** _____

Appendix C

Any supplemental material you wish to include (additional figures, tables, snippets of code, etc.).